GABRIEL JOHN SOLOMON

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EDUCATION

University of the East (UE) - Caloocan Aug 2023 - Present (Expected Jun 2027)
Bachelor of Science (BS) in Computer Science
GPA: 1.04 / 1.00 (Highest) - Latin Honor Standing, University Full Scholar

Awards: Technify 2026 - Best Research Paper (Computer Science), Philippine Junior Data Science Challenge 2024 - Top 10 Finalist, CodeChum National Programming Challenge 2025 - Semifinalist, Caffeine AI Manila Hackathon - Judge and Mentor, Devcon Game Jam - Top 8, AWS Community Day Manila - Event Co-Organizer

Leadership: Association of Computer Studies Students - UE Caloocan (President), Google Developer Student Club - UE Caloocan (Data Science Lead), AWS Learning Club - UE Caloocan (Vice-Chief Operations Officer), AWS Cloud Club Philippines (Data Analyst), AWS User Group e:Novators Philippines (Ambassador), Jesus Is Lord Caloocan (Technical Support)

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL (MySQL, PostgreSQL), HTML, CSS

Machine Learning & Computer Vision: PyTorch, TensorFlow, Scikit-Learn, Ultralytics YOLOv8, Segment Anything (SAM), OpenCV, Roboflow, scikit-fuzzy

Generative AI & LLM: LangChain, Hugging Face, RAG, ChromaDB, Pydantic, Ollama (local LLMs), Claude Agent SDK, Claude Code, Gemini, Groq

MLOps & Infrastructure: FastAPI, Next.js, Supabase, Microsoft Azure (Linux VMs), AWS, Elasticsearch, n8n, Git/GitHub, Vitest, Playwright

Data & Productivity: Pandas, NumPy, Matplotlib, Plotly, Power BI, Power Automate, Jupyter, Google Colab, Kaggle, VS Code, Microsoft Office

PROFESSIONAL EXPERIENCE

RivanAI Cybersecurity Inc. 2026
AI Security Intern | Project-based Program

- Engineered and executed **advanced adversarial simulations**, including direct and indirect prompt injection attacks and data poisoning vectors, to proactively identify vulnerabilities in LLM applications.
- Supported the secure development and deployment of Large Language Models (LLMs), implementing technical safeguards and **mitigating adversarial risks** across data pipelines, training phases, and inference layers within hybrid infrastructure.
- Built **controlled ransomware-behavior simulations** for SOC training, modeling real-world social engineering and extortion campaigns to evaluate and strengthen defensive posture and incident response readiness.

PROJECTS

OASYS - NeuroSymbolic Road Defect Detection & Severity Assessment Feb 2026 - Mar 2026

- Developed a **neurosymbolic pipeline** fusing YOLOv8 instance segmentation (Ultralytics, Roboflow) with a symbolic **Mamdani fuzzy engine** that grades pothole and crack severity against the 3 mm DPWH threshold, with an auditable reasoning trace per grade.
- Achieved a Bounding Box **mAP@0.5 of 98.7%** and Mask **mAP@0.5 of 98.3%**, validated over 100 epochs with Precision-Recall curves and confusion matrices to minimize background false positives across all defect classes.
- Built the end-to-end system: road detection, inverse perspective mapping, **Segment Anything** masking, skeletonization with GSD-based real-world measurements, privacy blurring, and a **local LLM (Ollama)** that drafts maintenance bulletins; served via FastAPI, Next.js, and Supabase.
- Research paper awarded **Best Research Paper (Computer Science)** at Technify 2026.

Cross-Encoder Reranking RAG Model Chatbot | RivanAI May 2026 - Jun 2026

- Developed a production-grade Retrieval-Augmented Generation (RAG) system implementing **LLM-based intelligent chunking, query rewriting, and cross-encoder reranking**.
- Built a native Python implementation integrating ChromaDB with HuggingFace embeddings (all-MiniLM-L6-v2), **custom Pydantic models** for structured output, and parallel processing for efficient document ingestion.
- Created vector space visualizations using **t-SNE dimensionality reduction (2D/3D)** with Plotly to analyze document clustering and validate embedding quality across multi-category knowledge bases.

GABO - Multi-Agent AI Harness 2026

- Built a web-based **multi-agent orchestration platform** on the Claude Agent SDK where 12 role-specialized agents hand off, challenge, and build on each other's work across four collaborative workflows (research, idea tournaments, hackathon build-and-ship, custom teams).
- Designed a **complexity-based router** that sizes agent teams without extra model calls, cutting token use by ~50% on quick tasks and reducing response latency from 8-9 s to ~3 s through warm session pooling.
- Enabled **multi-provider inference** (Claude, OpenAI, Gemini via a built-in translator, and local models via Ollama) with token-budget modes and context-window guards; tested with Vitest and Playwright.

n8n Automated Lead Orchestration Pipeline Jun 2026

- Architected an **automated CRM pipeline** in n8n that ingests inquiries via webhooks, reducing initial response time from hours to under 2 minutes.
- Integrated a **Groq Chat Model** for semantic analysis of unstructured text, extracting service categories, scoring lead sentiment, and routing leads via custom JavaScript modules and Switch logic.
- Automated document generation through the Google Drive API and built an **asynchronous error-handling framework** that logs API timeouts and dispatches instant alerts, ensuring 100% data capture.

Cloud-Based Security Incident & Ticketing Platform | RivanAI May 2026 - Jun 2026

- **Deployed** an OTRS (Open-source Ticket Request System) environment on **Azure Linux** to centralize incident tracking and streamline security/IT workflows.
- **Integrated Elasticsearch** for high-performance querying, accelerating log analysis, search, and event triage.
- **Configured and hardened** the cloud infrastructure for a secure, scalable, and highly available service management pipeline.

CERTIFICATIONS

- AI Engineer for Data Scientists Associate
- Certified AI Agent Security Specialist
- Certified AI Data Security Specialist
- AI Fluency: Framework & Foundations
- Introduction to Machine Learning
- Introduction to Data Science
- Introduction to Python
- Introduction to Programming

CHARACTER REFERENCE

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